0B.M.S. COLLEGE OF ENGINEERING BENGALURU
Autonomous Institute, Affiliated to VTU

Lab Record

Object Oriented Analysis and Design

Submitted in partial fulfillment for the 5th Semester Laboratory

Bachelor of Technology
in
Computer Science and Engineering

Submitted by:

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B.M.S. COLLEGE OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Object-Oriented Analysis and Design(23CS6PCSEO) laboratory has been carried out by G M Kusuma (1BM24CS405) during the 5th Semester Aug-Dec-2025.

Signature of the Faculty Incharge:

NAME OF THE FACULTY: RoopaShree CS

Department of Computer Science and Engineering
B.M.S. College of Engineering, Bangalore

Table of Contents

1. Hotel Management System

SRS DOCUMENT

Software Requirement Specification on
Hotel Management System

1. Introduction

1.1 purpose

This document specifies the Software requirement for the Hotel Management System. The system will automate hotel operations such as room booking, check in/check-out, billing and reporting to improve operational efficiency and guest experience

1.2 Document Conventions

-> Bold for heading and imp terms
-> Bullet Point for list
-> use clear, simple language.

1.3 Intended Audience and Reading Suggestion.

-> Developers and testers for implementation and validation.
-> project managers for planning and tracking
-> Hotel staff and stakeholders for understanding capabilities.

1.4 product scope

HMS is a web-based application designed to manage hotel reservation guest checkin/checkout payment, user roles and reporting.

1.5 References

-> IEEE Std.
-> PCI DSS guidelines for payment security.
-> GDPR for data protection compliance.

2. Overall Description

2.1 product perspective

The HMS is a standalone web application integrated with payment gateways and notification services.

2.2 product functions

-> user registration and authentication.
-> Room search and booking
-> booking notification and cancellation
-> Guest check-in and check-out
-> payment processing and invoicing
-> Report generation.

2.3 user classes and characteristics

Admin: full access, manages sys and reports
Receptionist: Manages bookings and guest operations
Guest: searches and books rooms, manages profile.

2.4 Operation Environment

-> modern web browsers on desktop and mobile
-> cloud or onpremise server desktop and mobile

2.5 Design and Implementation Constraints

-> secure payment processing
-> Data encryption in transit and at rest
-> Role based access control

2.6 User documentations

-> user manuals and online help
-> Admin and staff training guides

2.7 Assumptions and Dependencies

-> Internet connection for payment and notification
-> Third party API's availability.

3. Specification Requirements

3.1 functional Requirements

-> Secure user Registration and login
-> Role based access control
-> Room availability search by date, type and price
-> Book, modify and cancel reservation
-> check in and check-out processing
-> process payments and generate receipts.
-> manage user profiles

→ Admin reports on bookings, revenue and Occupancy
→ Notification via email/SMS

3.2 External Interface Requirements
→ User-friendly web UI
→ payment gateway integration (Stripe, Paypal)
→ Email and SMS notification API's
→ Secure HTTPS Communication

3.3 System features
→ Room management: add/update/delete rooms and availability
→ Booking management: full Booking lifecycle handling
→ User Management: Roles and access controls.
→ payment processing and Billing
→ Reporting Dash board.

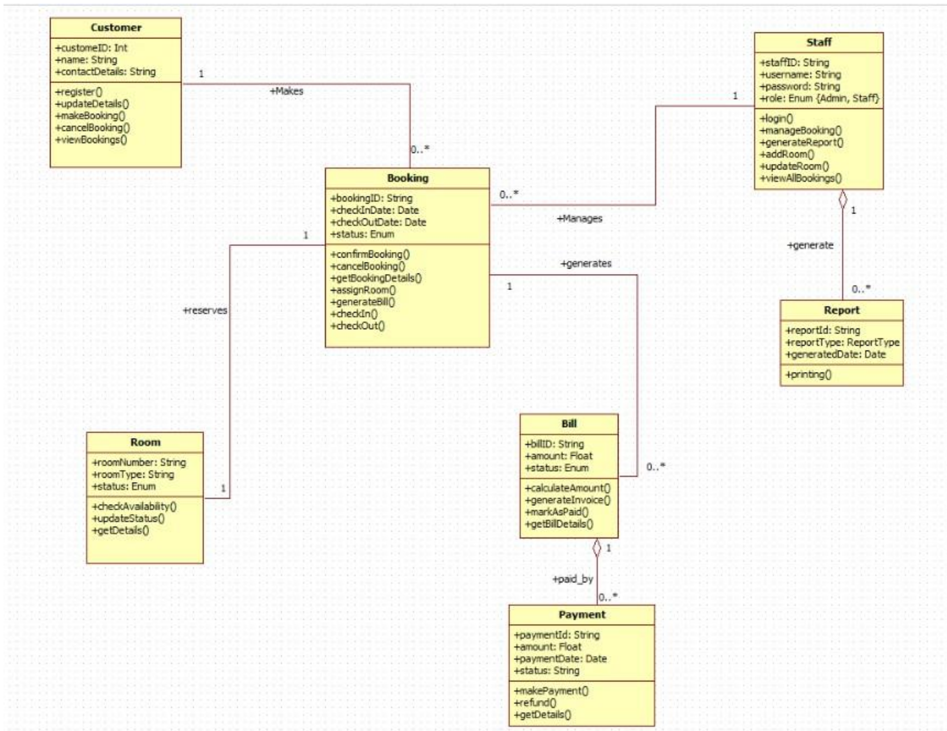
3.4 Non-functional Requirements
→ Support 100+ Concurrent user
→ Response time under 2 seconds
→ Data security and encryption.
→ 99.9% system uptime
→ Scalable architecture for future Expansion

4. Appendices

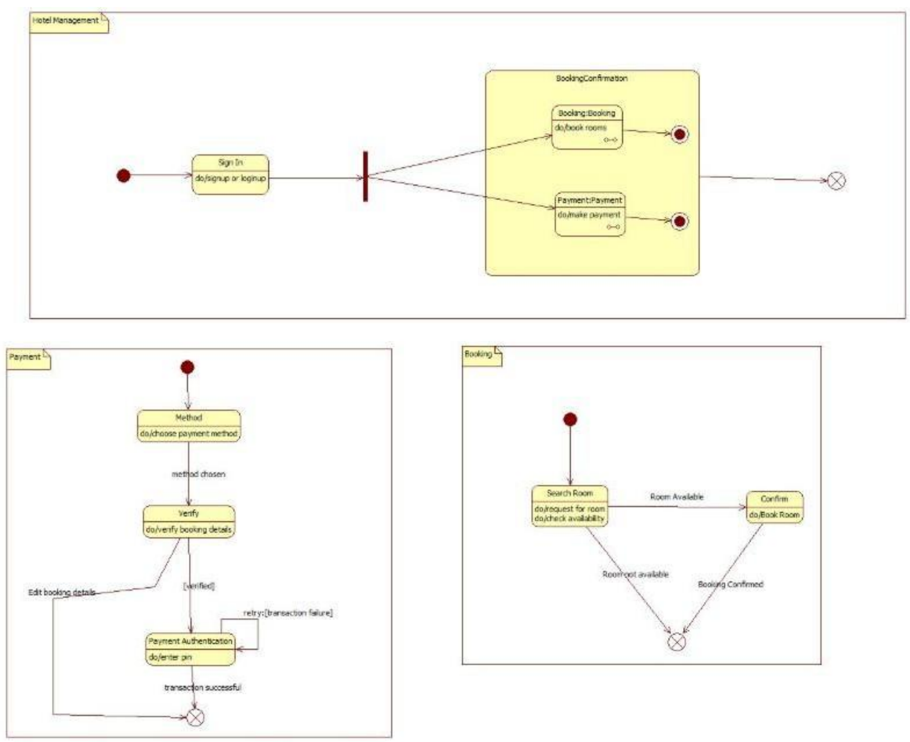
4.1 Glossary
→ Booking: Reservations of Rooms
→ check in: Guest arrival process
→ check out: Guest departure and payment finalization.

4.2 future enhancements
→ Mobile apps
→ Door lock system
→ loyalty programs.
→ Multi-language support.

CLASS DIAGRAM

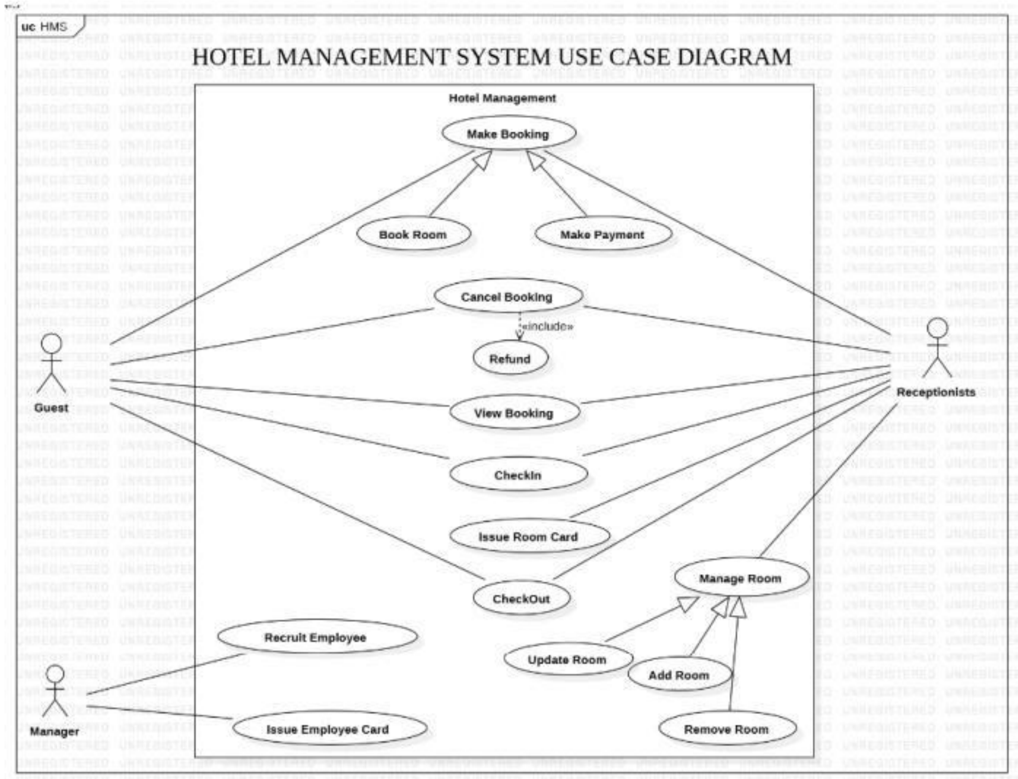


STATE DIAGRAM

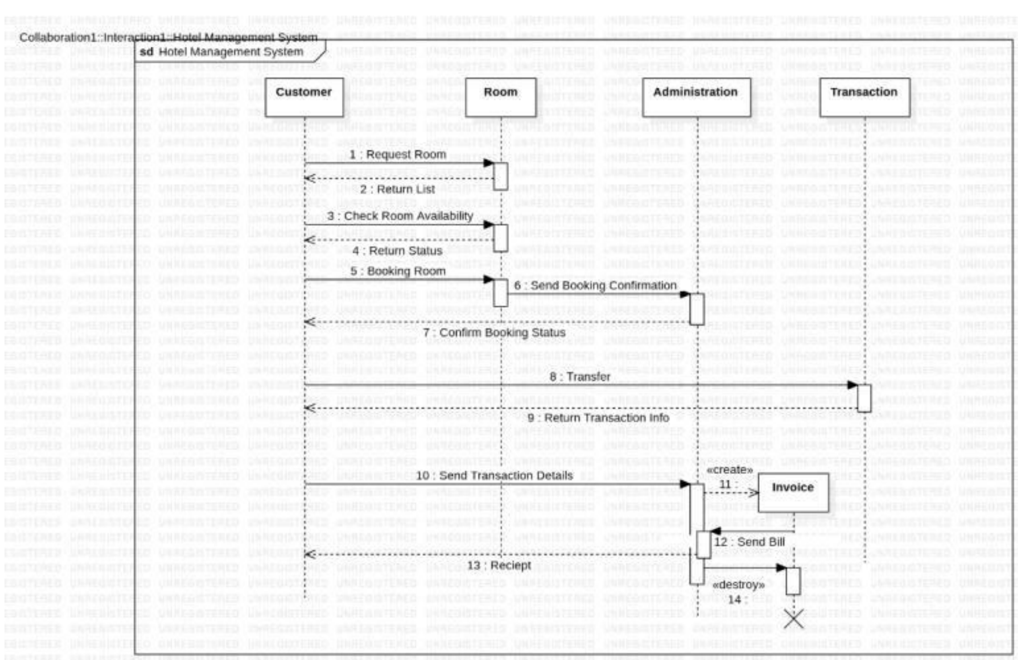


INTERACTION MODELS

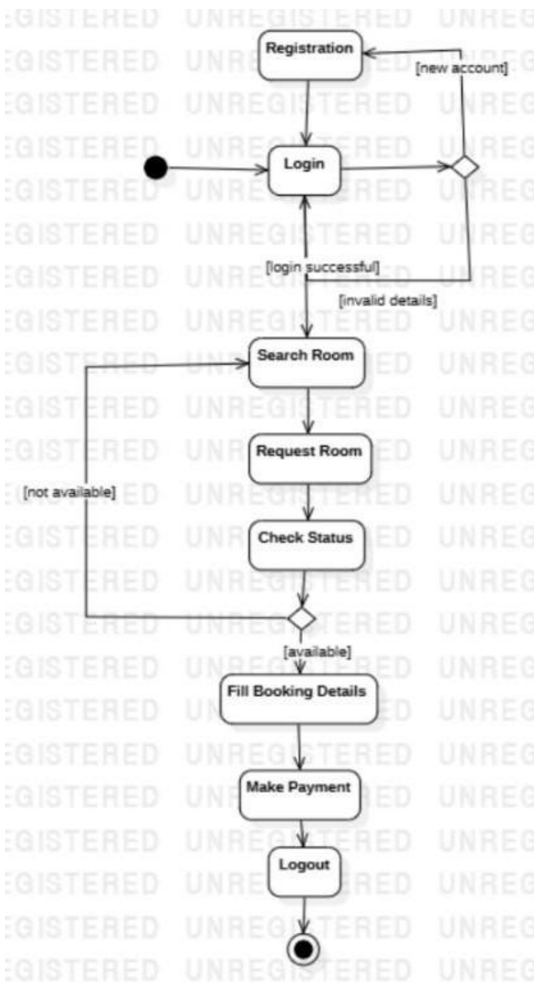
a. Use case model



b. Sequence model



c. Activity model



2.Credit Card Processing

2.1 SRS DOCUMENT

Software Requirements specification for Credit card processing System

1. Introduction

1.1 purpose

It is a secure system designed to authorize, process and settle credit card transactions for merchants. This SRS covers the core transaction processing functionalities including authorization, capture, refund and settlement.

1.2 Document Conventions

-> Requirements are standalone and do not inherit priority from other requirements.

-> IEEE format

1.3 Intended Audience and Reading Suggestions

-> This SRS is intended for developers, project managers, testers and stakeholders. Readers are encouraged to start with the specific requirements based on their roles.

1.4 project scope

-> It supports authorization, settlements, refunds handling and fraud detection. The system aligns with organizational goals to modernize and secure transaction handling.

1.5 References

Credit card network API guides

2 Overall Description

2.1 product perspective

-> It is a standalone system replacing older payments system. It interfaces with card networks, banking API's. The system supports integration with external services through standardized APIs

2.2 product functions

The system authorizes transactions, captures payments, process settlements, handles chargeback and generate reports

2.3 user classes and characteristics

user groups includes customer, merchants, admin staff and developers.

2.4 Operating Environment

The system runs on a cloud platform with database. 2.0 for secure Communication.

2.6 user documentation

-> will be provided in both formats

It will be integrated in the user interface

2.7 Assumptions and Dependencies

-> The system depends on external card networks, stable API'S, and ongoing access to cloud info. It assumes user access the system through secure channels.

3 Specification Requirements

3.1 functional requirements

-> Authorize transaction within 3 sec

-> capture payment after merchant confirms

-> process merchant - initiated refunds.

-> keep transaction logs for 2 years.

3.2 External Interface Requirement

-> use standard version for payment network messaging

-> provide Restful APIs for merchant

3.3 External System features

-> Role based access control

-> Real time monitoring dashboard

-> Automated backup and recovery.

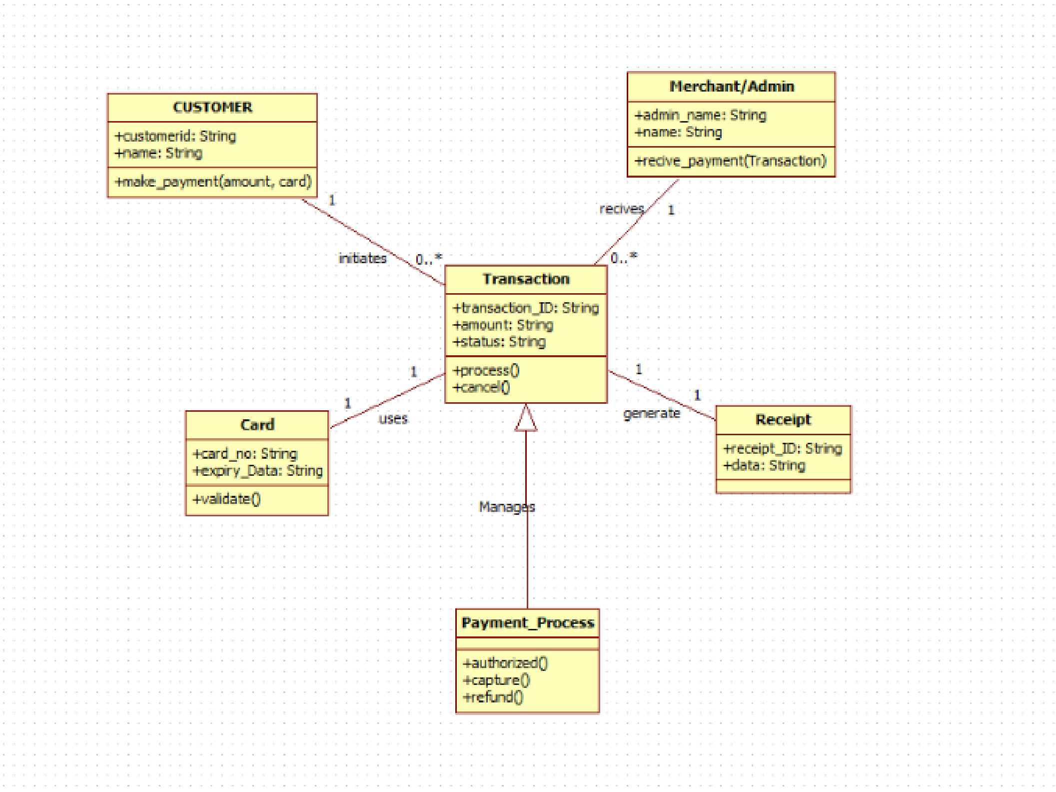
3.4 Non-functional Requirement

-> 99.9% system availability

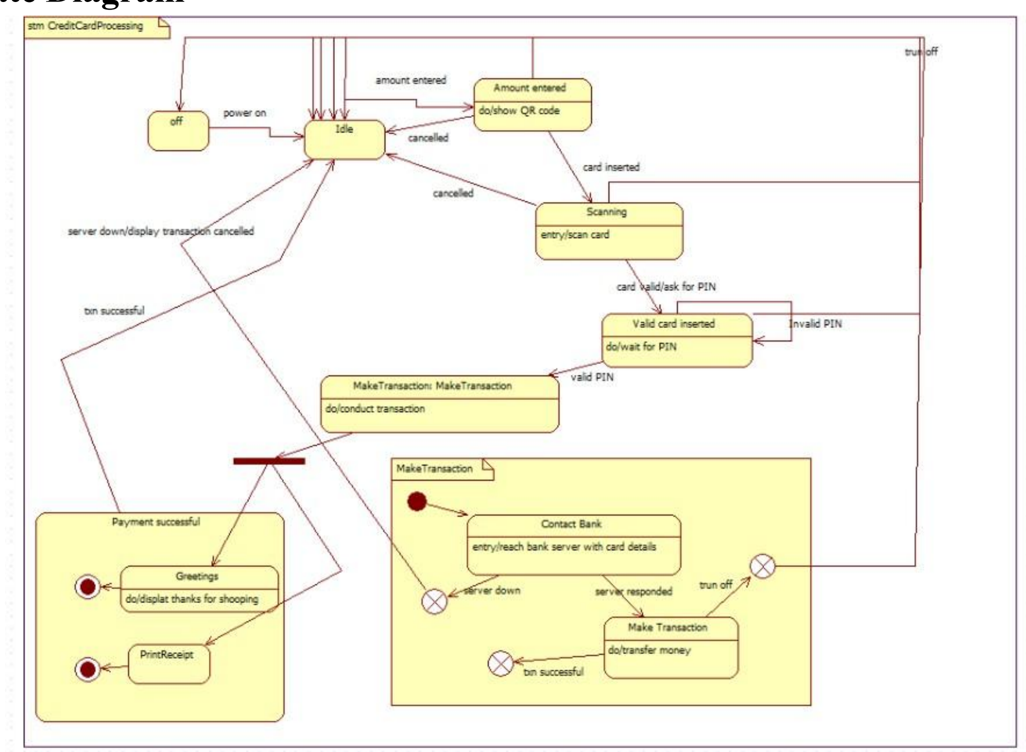
-> Encrypt all sensitive data.

-> Handles 10,000 transaction /sec

2.3 Class Diagram

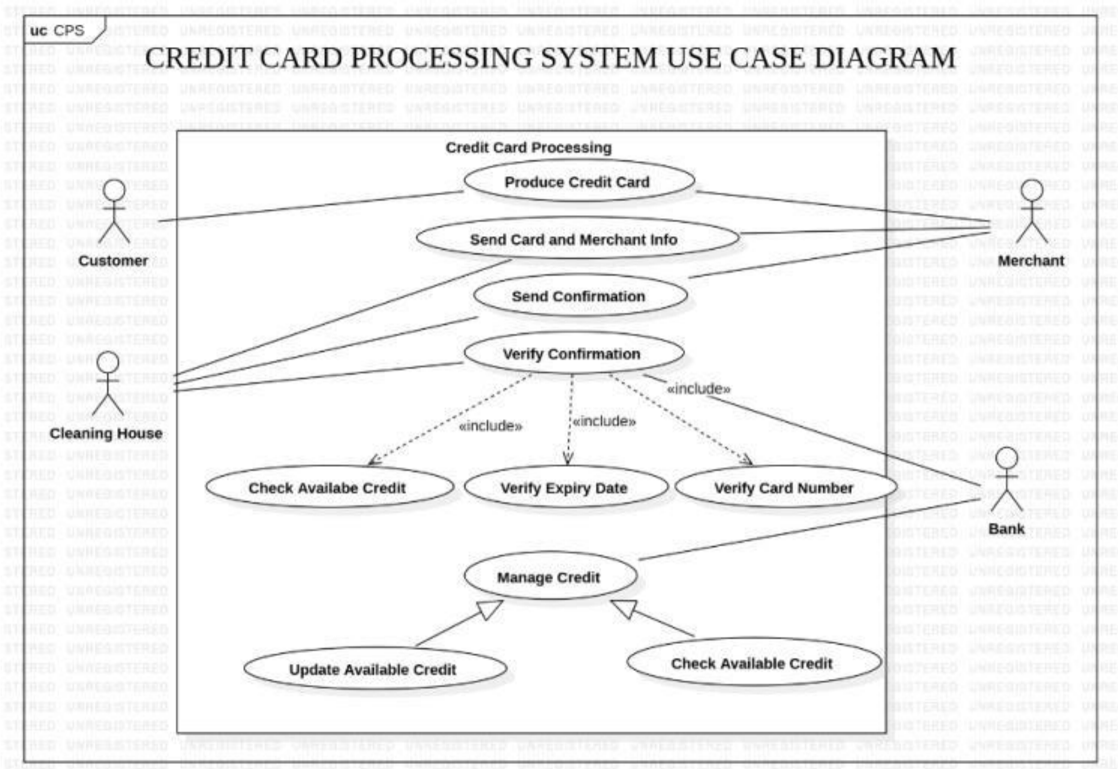


2.3 State Diagram

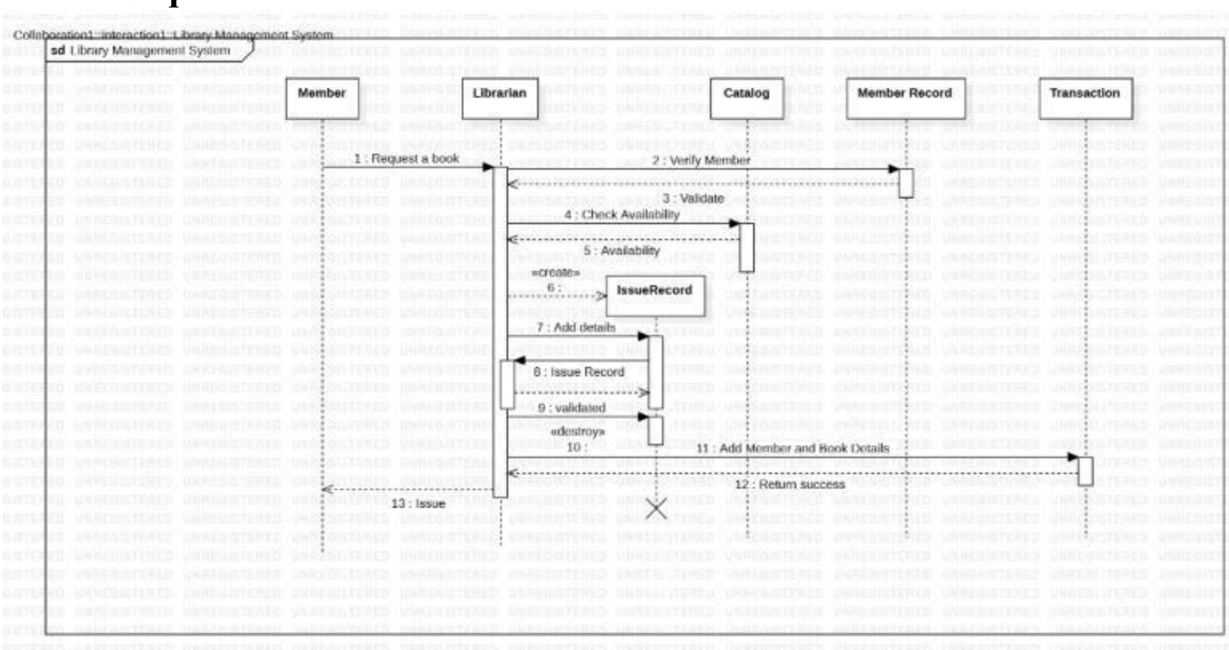


2.4 Interaction Models

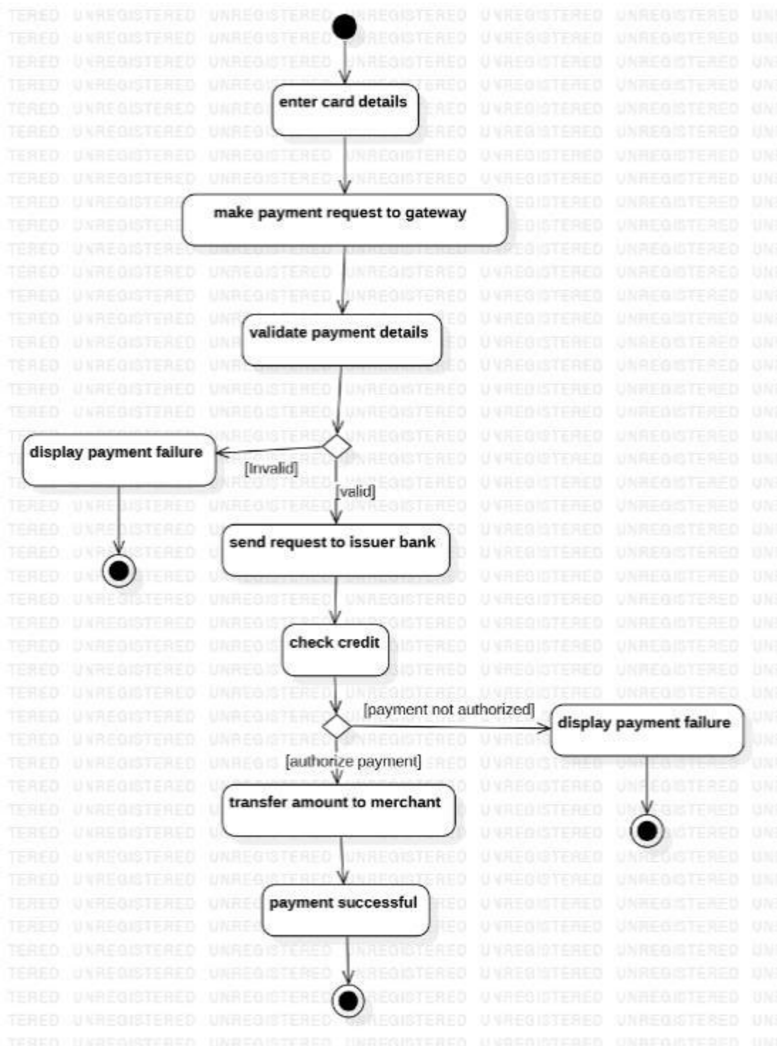
a. Use Case Model



d. Sequence model



c. Activity Model



3.Library management system

3.1 SRS Document

Library Management System

1. Introduction

1.1 purpose

The purpose of this doc is to Specify the requirements for the library Management System (LMS) which automates the process of managing books, members, transactions

1.2 Document Convention

The document follows IEEE standards for S/w Requirements Specification.

1.3 Intended Audience

This doc is for Developers, testers, librarians & project stakeholders.

1.4 project Scope

The LMS provide functionalities for book cataloging, members registers, issuing, returning & fine calculation. It improves efficiency & reduces manual errors.

1.5 References

IEEE SRS format, standards database design practices & sample library workflows

2. Overall Description

2.1 product perspective

The systems Replaces manual registers with a centralized computerized database.

2.2 product function

It manages book records, user membership, transaction & report generations.

2.3 Operating Environment

web based application running on windows/linux server with a SQL Database.

2.4 Constraints

Access restricted by roles: transaction must be logged

2.6 Documentation

user manual and admin guides will be provide

3. Specific Requirements

3.1 function Requirements

-> Add, update & delete book records.

-> Register & manage members

-> Issue and Return books with due-date tracking

-> calculate and manage fines.

3.2 External Interface Requirements

-> web interface for users and admin

-> Database connectivity

3.3 System Features

-> Search books by title, author or category.

-> Generate reports on usage & fines.

3.4 Non-functional Requirements

-> Secure Authentication

-> Response time < 2 Seconds

-> Scalable to handle 10,000 records.

4 Appendices

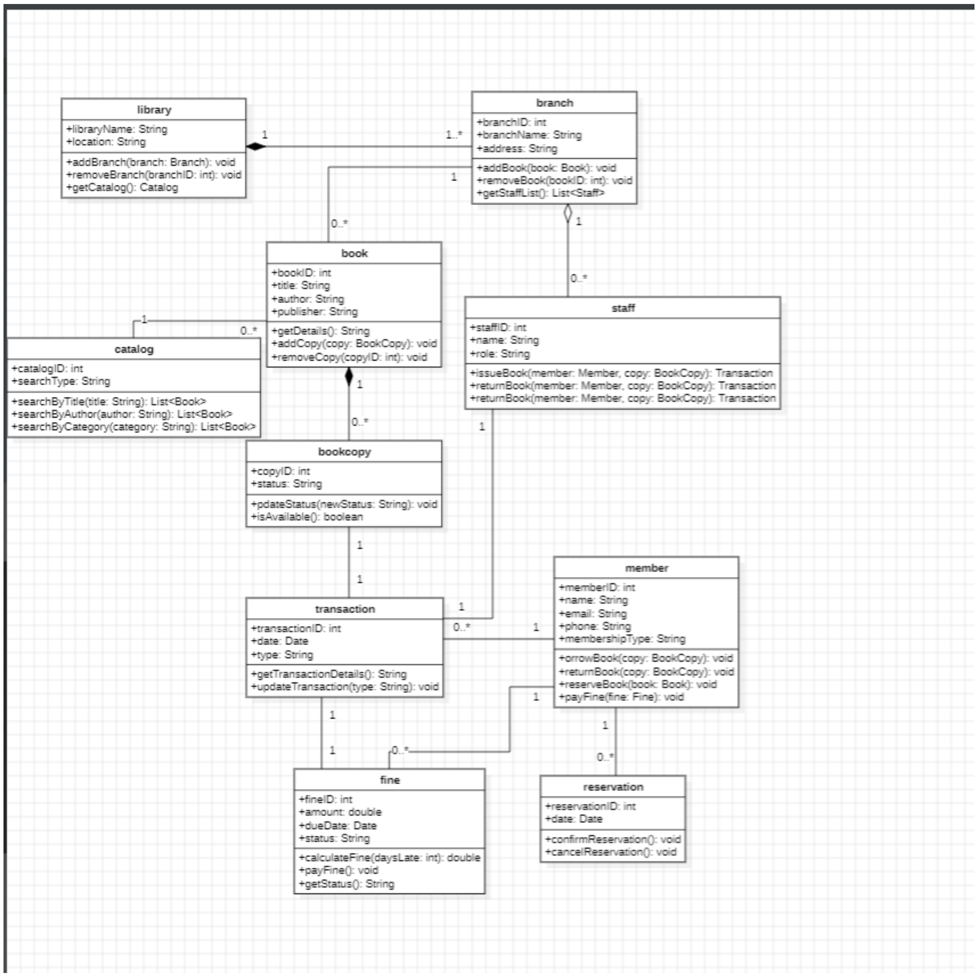
4.1 Glossary

LMS -> library managment sys, DB.

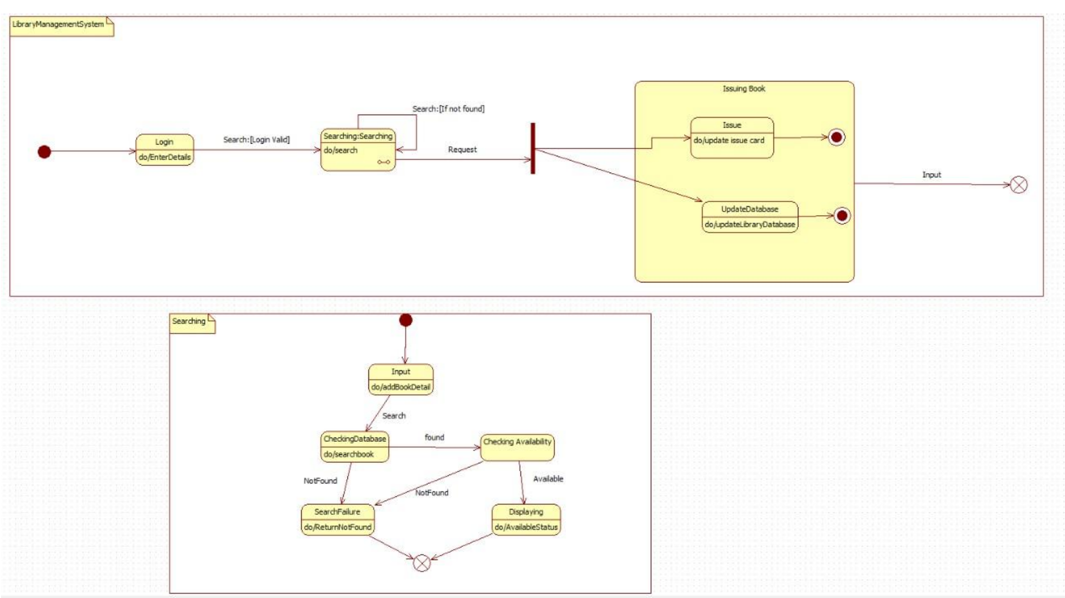
4.2 future Enhancement

Integration with e-books & digital library services.

3.3 Class Model

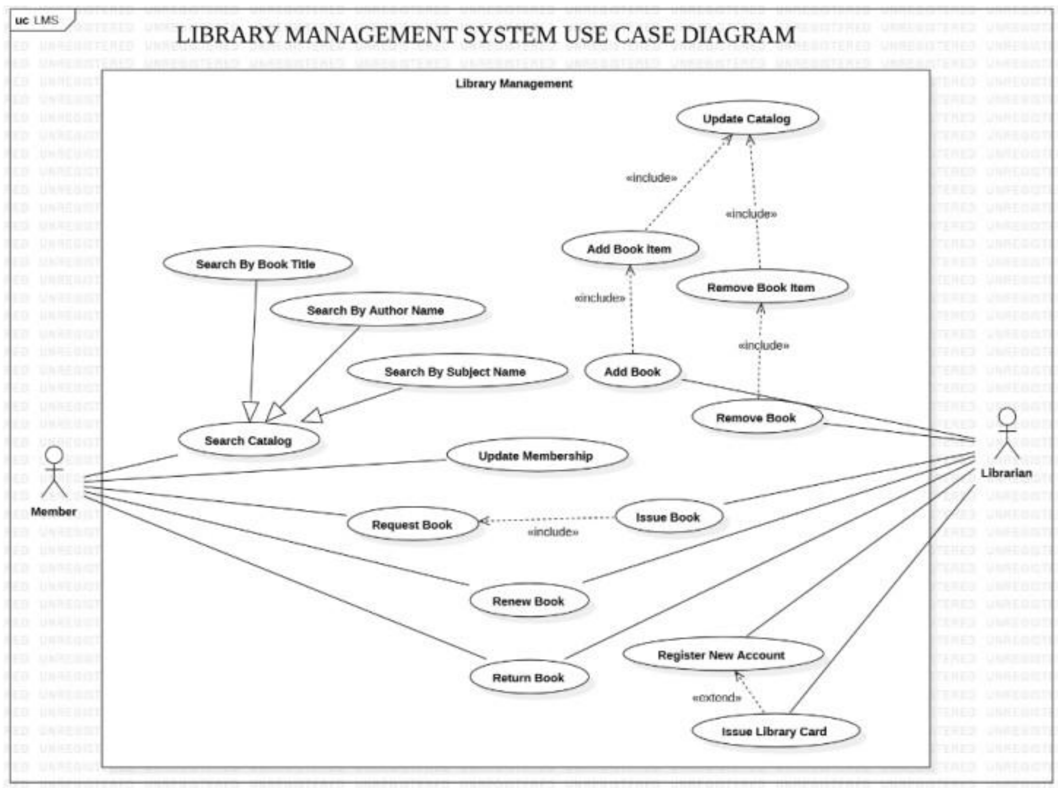


3.4 State model

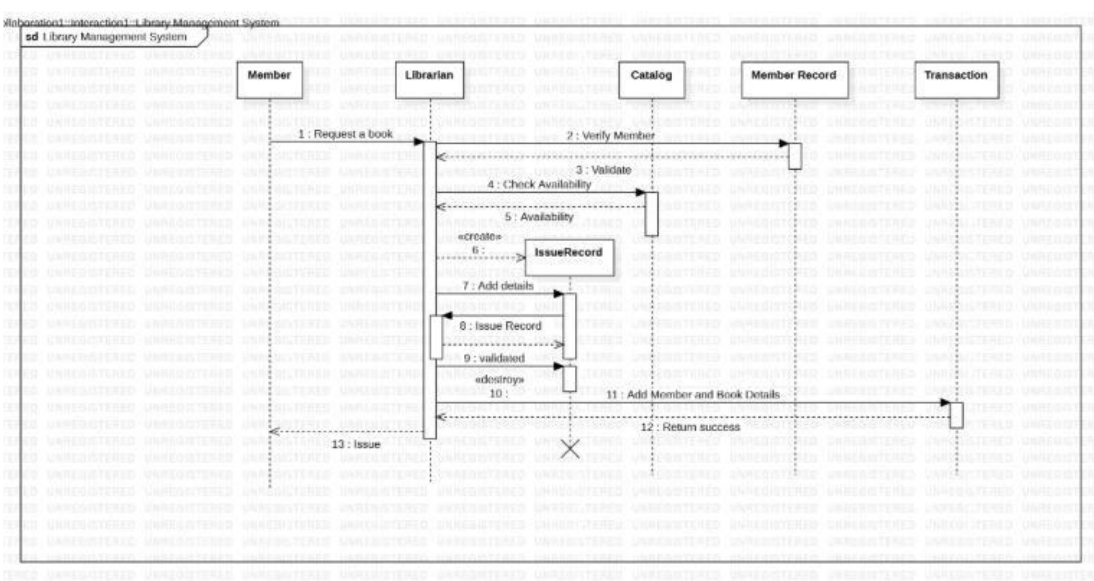


3.5 Interaction Models

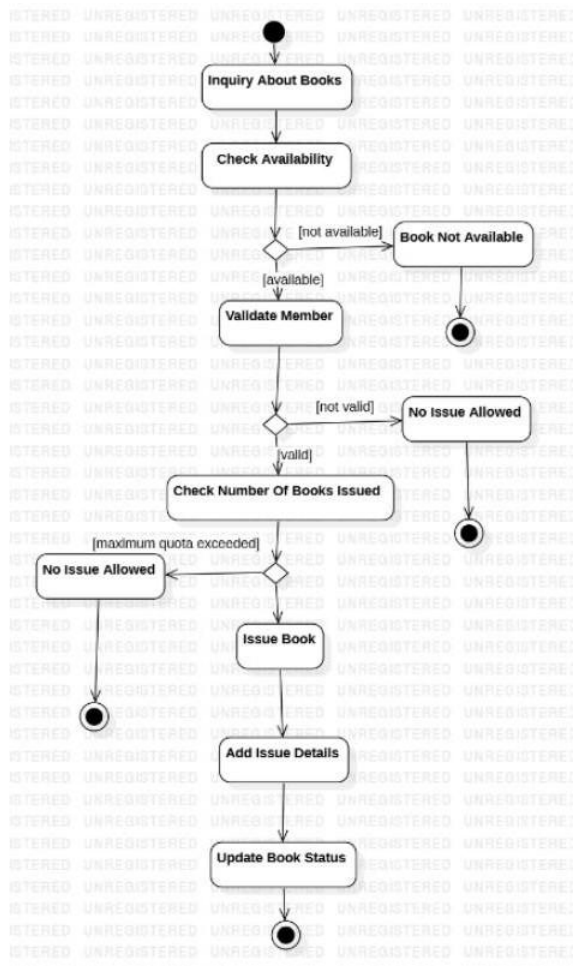
a. Use Case Model



b. Sequence Model



c. Activity Model



4.Stock Maintenance System

4.1 SRS Document

Stock Maintenance System

1. Introduction

1.1 purpose

The Stock maintenance sys (SMS) maintains Inventory details and streamlines stock tracking

1.2 Document Convention

This document to standard SRS structure

1.3 Intended Audience

The sys is for warehouse managers, sales, staff and developers

1.4 project scope

The system reduces manual stock handling, prevents shortages and ensures timely updates of inventory.

1.5 Reference

Inventory management best practices, IEEE SRS guidelines

2. Overall Description

2.1 product perspective

The system provides real-time stock management integrated with a central database.

2.2 product function

Stock Entry, update, reorder alerts & report generation

2.3 Operating Environment

Desktop/web application with database backend.

2.4 Constraints

system must handle concurrent access.

3. Specific Requirements

3.1 functional Requirements

-> Add and update stock items.

-> Track stock levels with alerts

-> Generate purchase orders when below

-> produce sales & inventory reports.

3.2 External Interface Requirements

-> Barcode Scanner Support

-> Database and reporting modules

3.3 System features

-> Real-time stock tracking.

-> low stock notification.

3.4 Non-functional Requirement

-> 24/7 availability.

-> Secure data storage.

-> Accuracy tolerance < 1%

4. Appendices

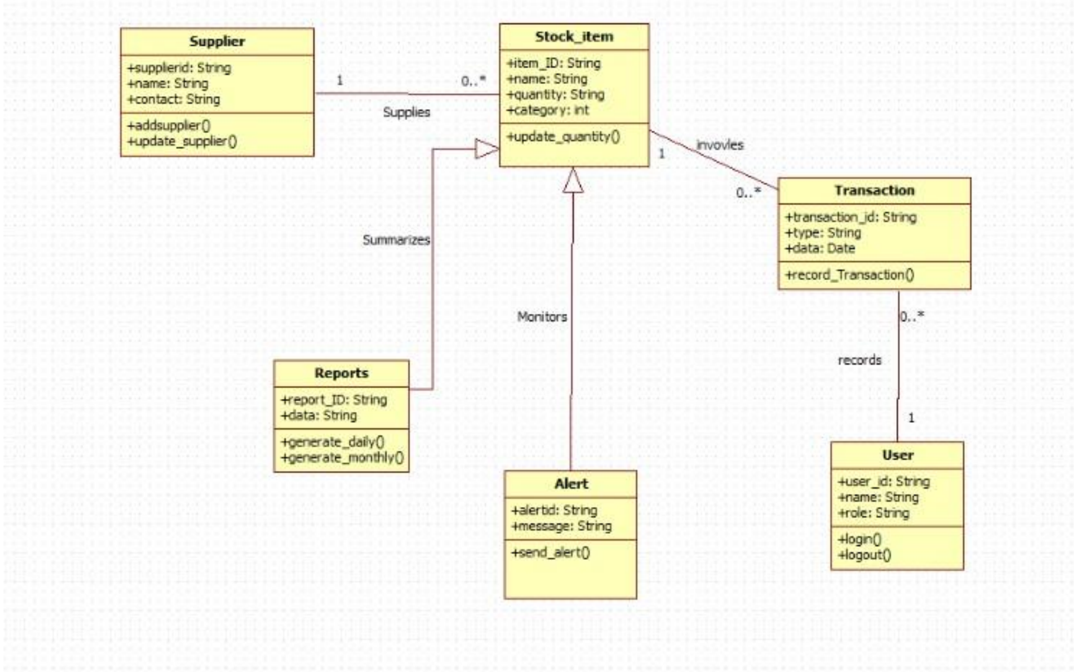
4.1 Glossary

SMS SKU - Stock Keeping unit.

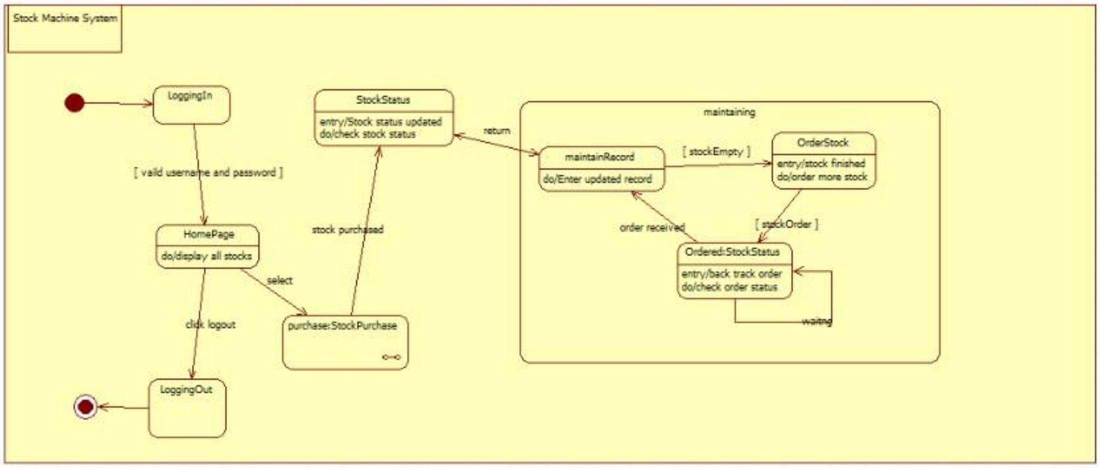
4.2 future Enhancement

Mobile app integration & AI-based demand forecasting.

2.3 class model



State Model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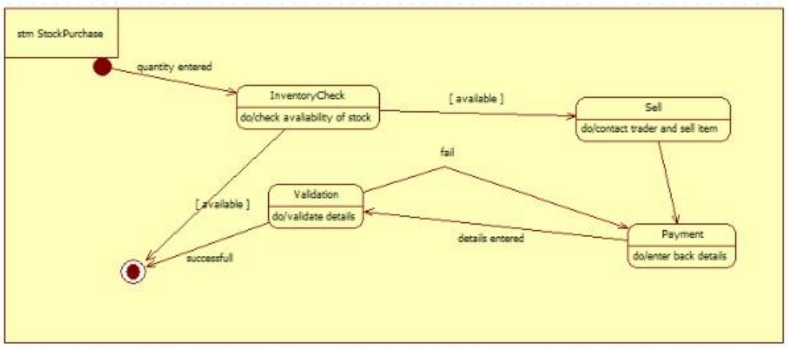
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fail
[available]
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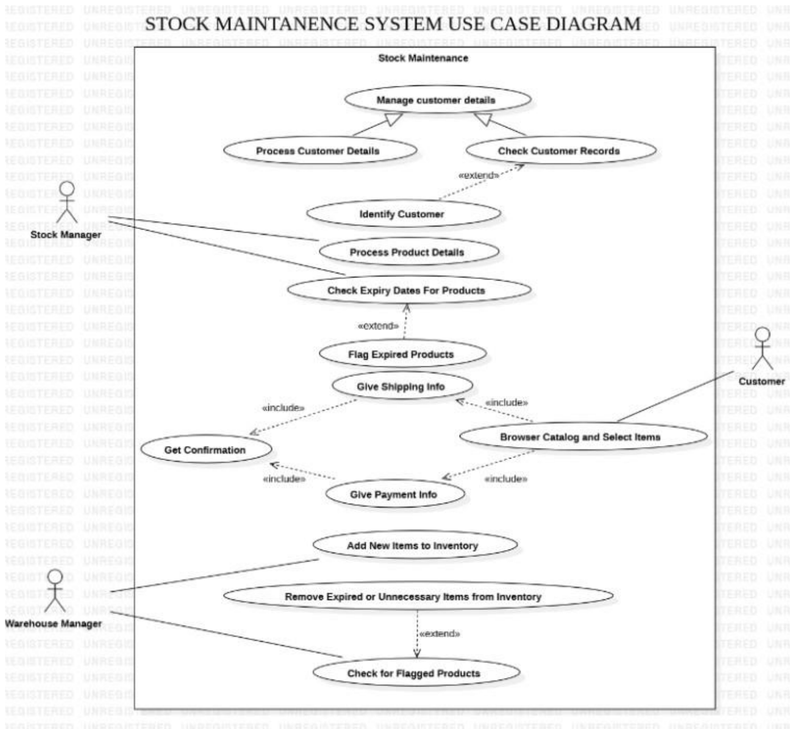
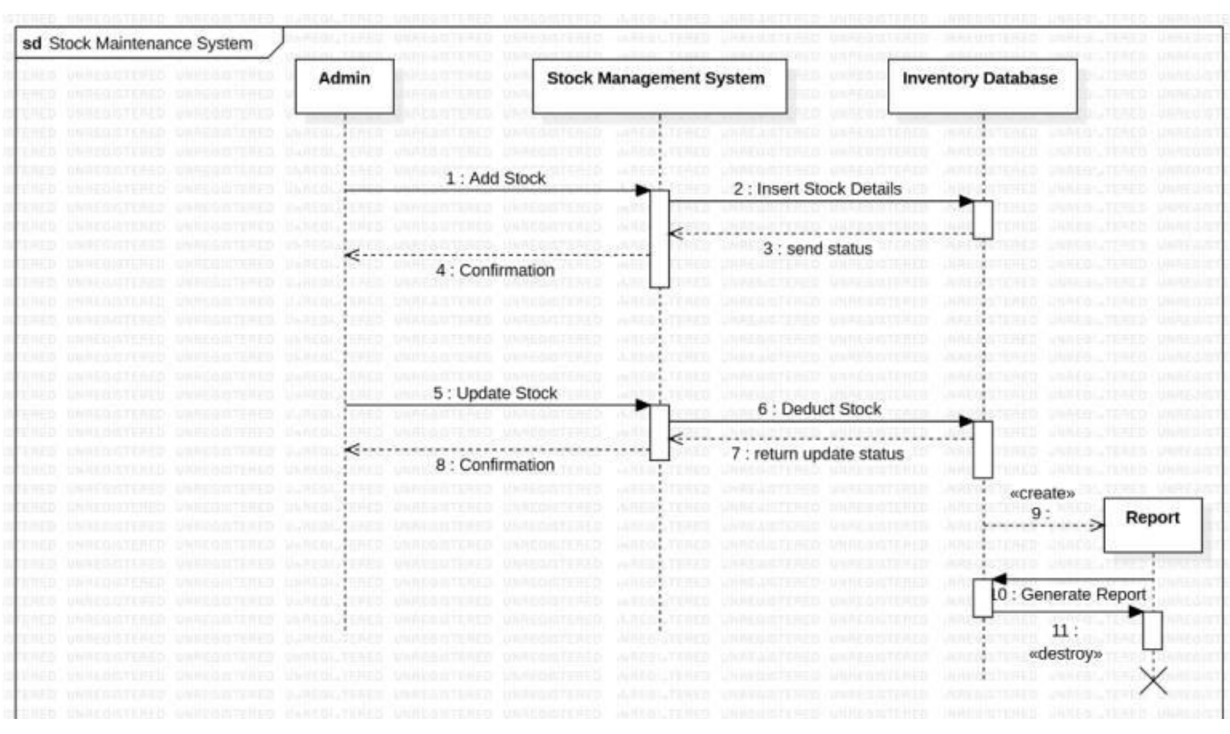
details entered
Payment
do/enter back details
successfull

4.3 Interaction Models

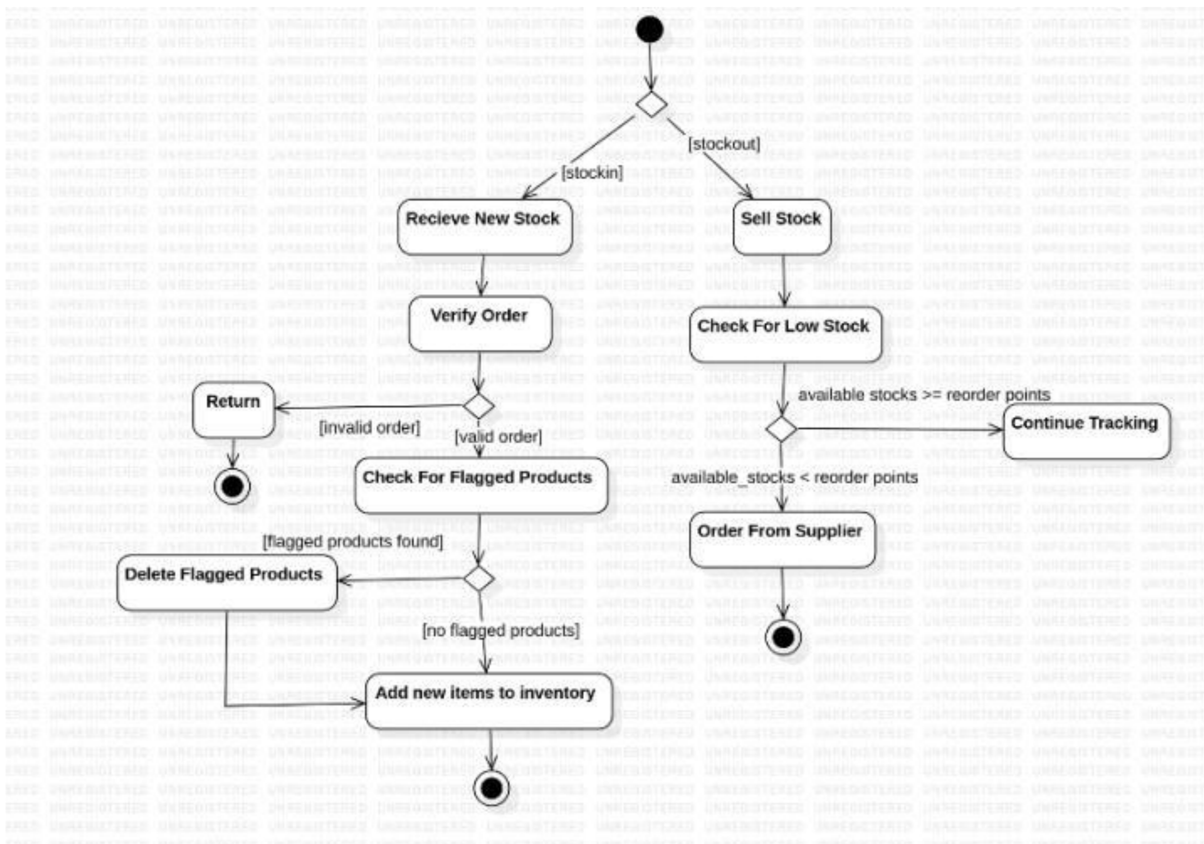
a. Use Case Model



d. Sequence Model



e. Activity Diagram



5 .Passport Automation System

5.1 SRS Document

Passport Automation System

1. Introduction

1.1 purpose

This document specifies requirements for an automated passport Automation System (PAS) that handles passport application & verification online.

1.2 Document Convection

IEEE style has been followed for clarity and consistency.

1.3 Intended Audience

Government officials, applicants & development teams.

1.4 project Scope

The system replaces manual passport processing ensuring faster application handling & tracking.

1.5 Reference

Govt passport issuance guide lines IEEE & SRS standards.

2. Overall Description

2.1 product perspective

The system provide an online platform connected to the govt database.

2.2 product function

Application submission verification, payment & appointment scheduling.

2.3 User classes

Applicants, Verification officers & administration.

2.4 Operating environment

web based system with secure authentication connected to national Database.

2.5 Constraints

Must comply with government regulations and data security policy.

2.6 Documentation

user guidelines and admin manuals will be supplied

2.7 Automation Assumptions and Dependencies.

Assume Connectivity with police & national ID Database.

3. Specific Requirements

3.1 functional Requirements

-> Submit online application
-> upload Required documents
-> Schedule appointments
-> Verify application details with police & govt records.
-> process online payments.

3.2 External Interface payment

-> web portal for users.
-> Integration with payment gateways & government databases.

3.3 Non-functional Requirement

-> High security for personal data
-> 99.9% uptime
-> fast response within 3 sec

4. Appendices

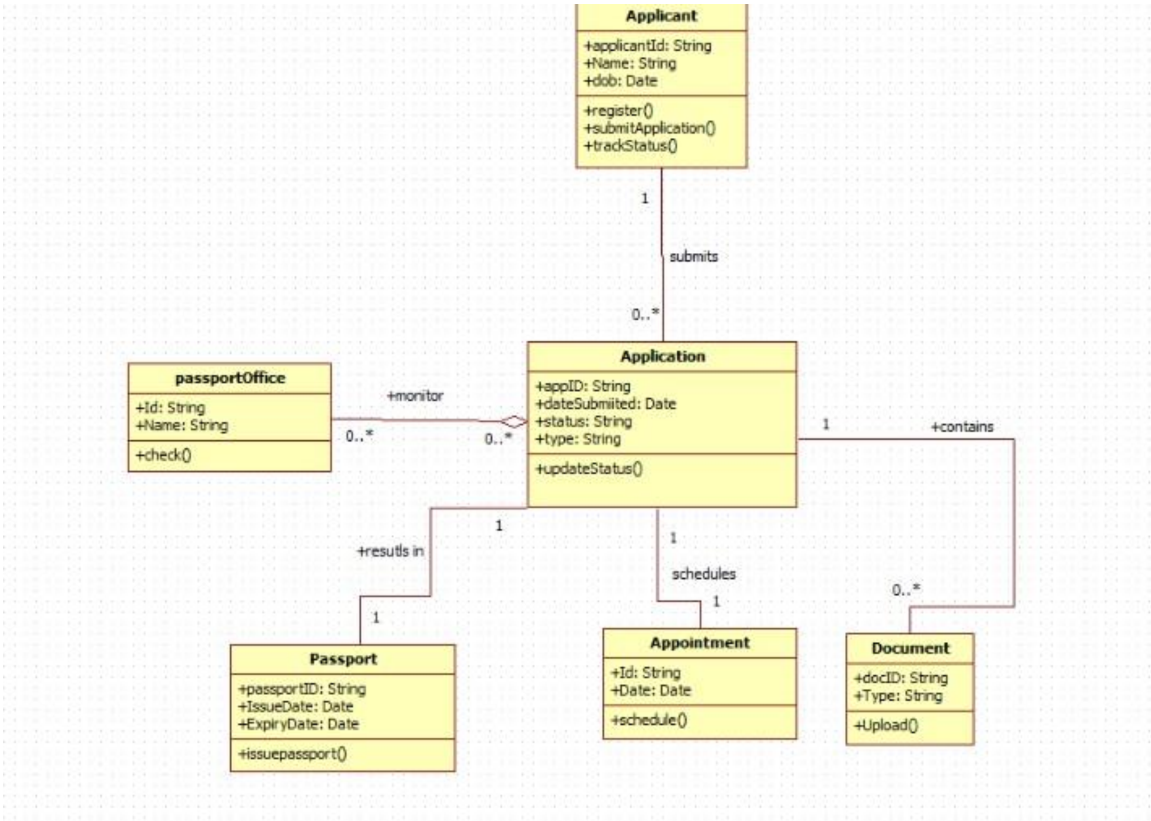
4.1 Glossary

-> PAS -> passport Automation System
-> ID -> Identification.

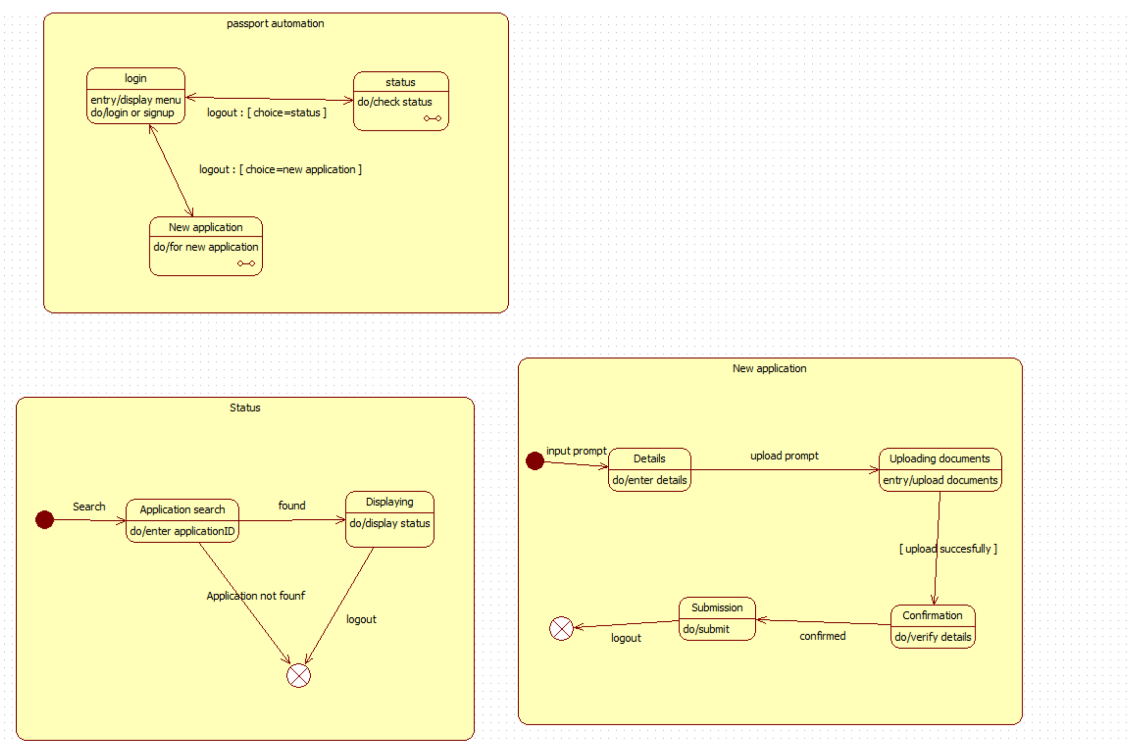
4.2 future Enhancement

-> Integration with Biometric systems & mobile application.

Class Model

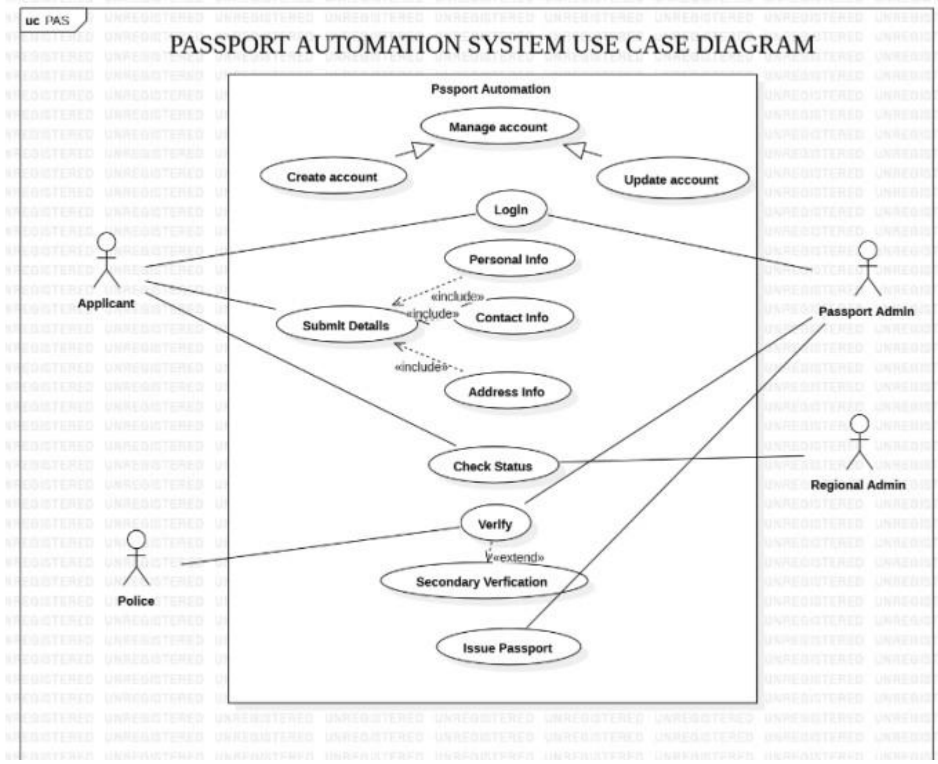


5.2 State Model

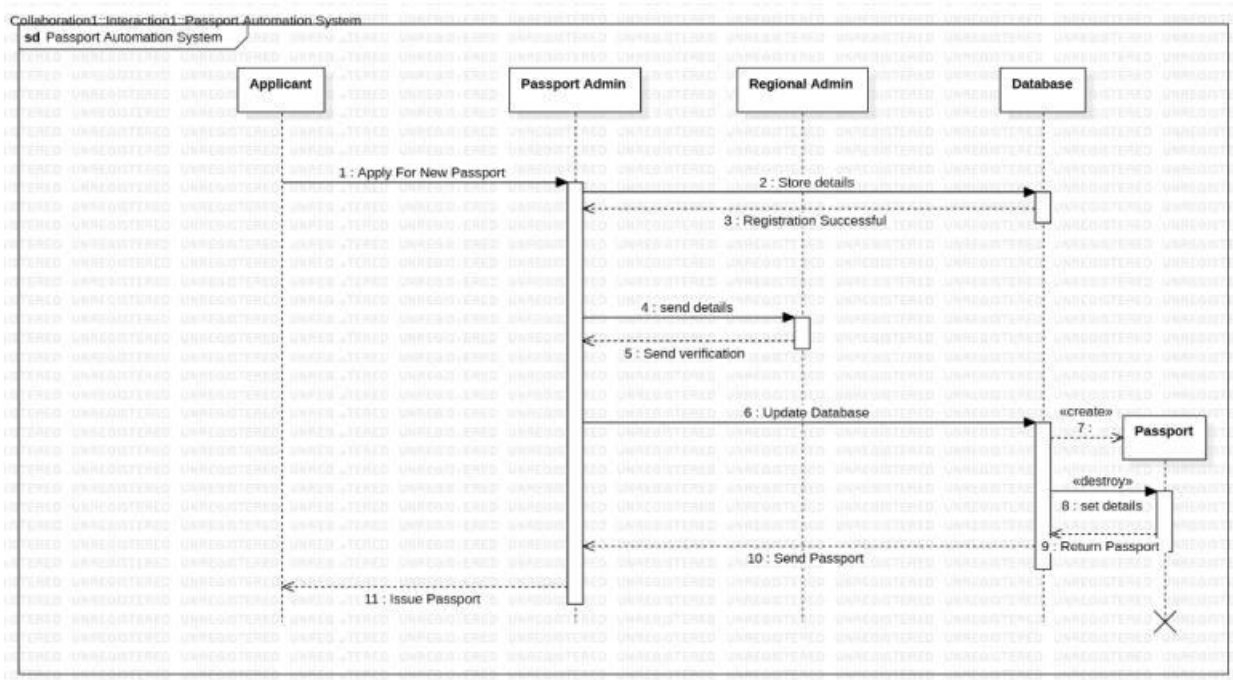


5.3 Interaction Models

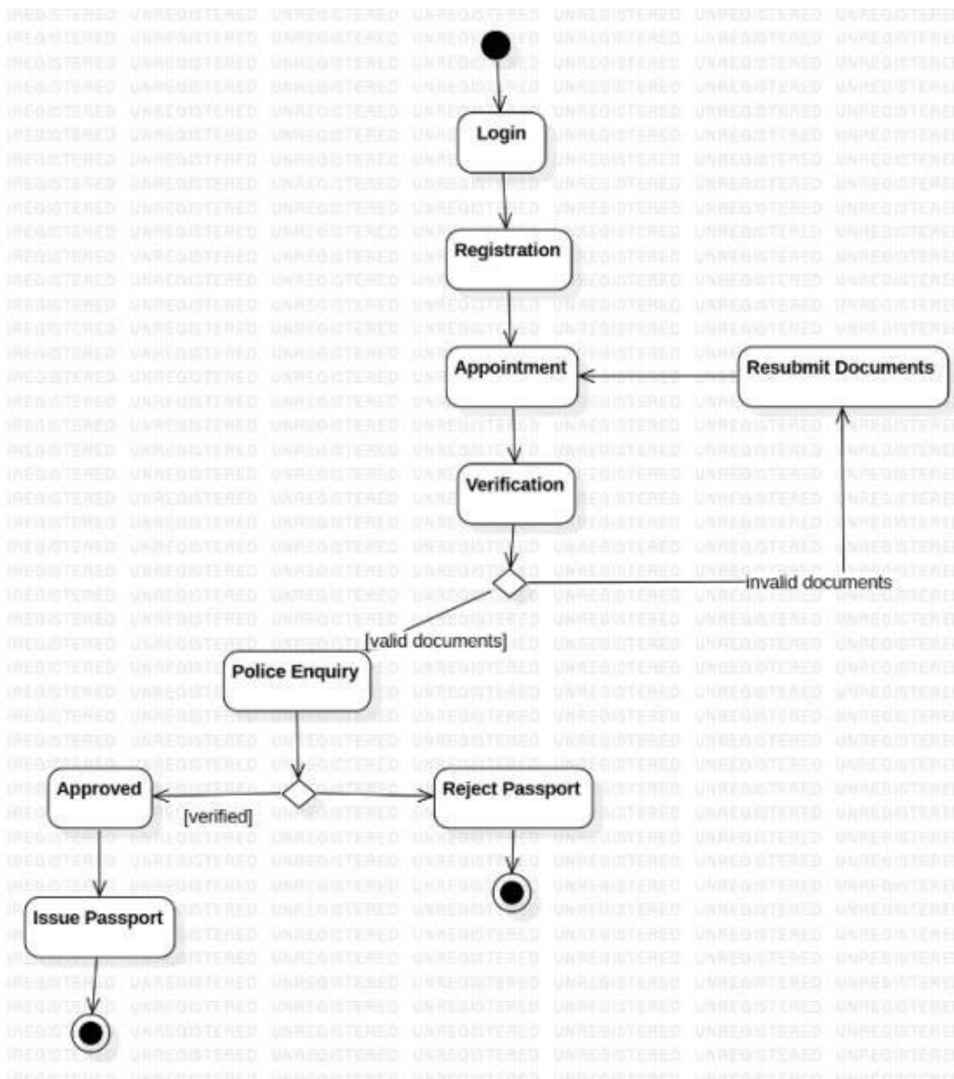
a. Use Case Model

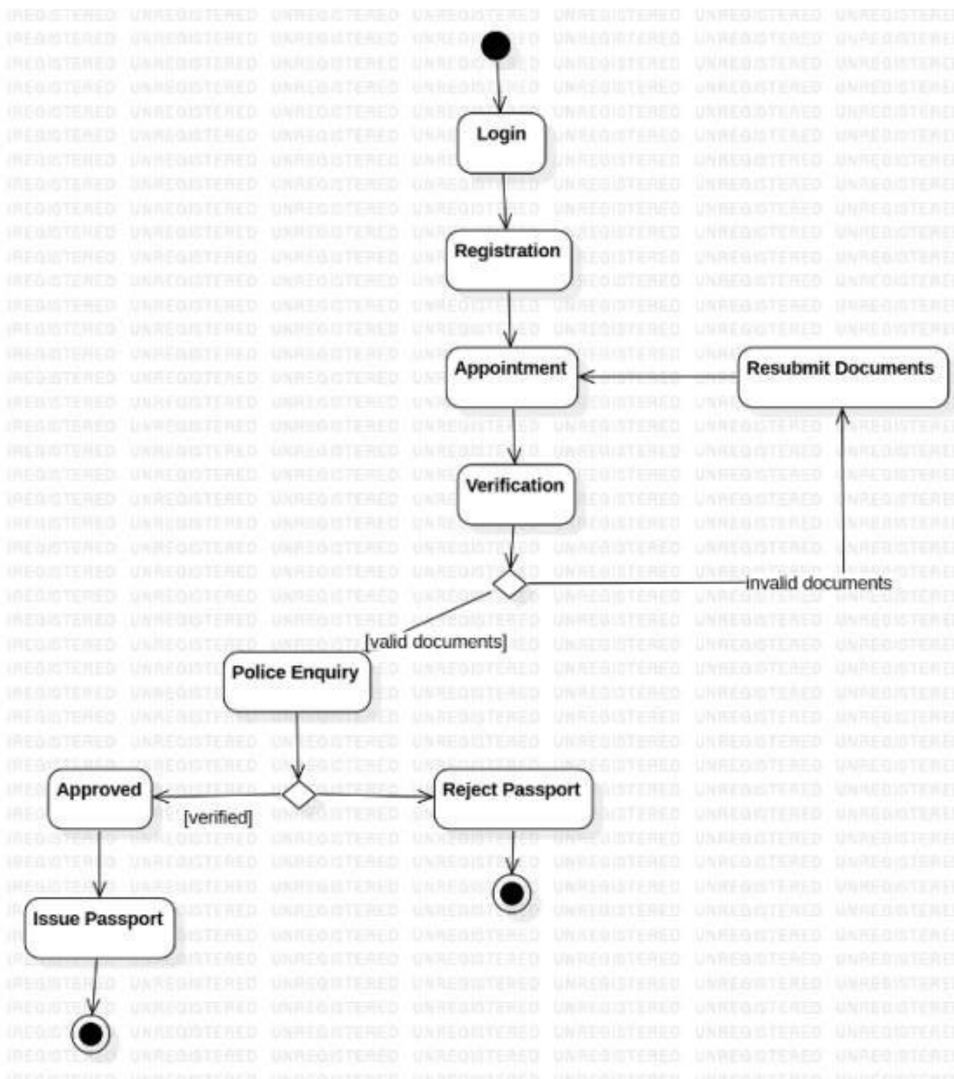


a. Sequence Model



b. Activity model




Login
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Appointment
Resubmit Documents
Verification
invalid documents
[valid documents]
Police Enquiry
Approved
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Reject Passport
Issue Passport